Vantage-Point-Aware Planning

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Abstract

Robots increasingly operate in visible public spaces, where they must perform sensitive tasks under the watchful eyes of bystanders. To maintain security and privacy, these machines must actively manage human observation, which occasionally requires them to strategically deceive onlookers. However, existing robotic planning approaches fail to model human vantage points or represent intentional deception within their frameworks. This limitation creates an urgent need for vantage-point-aware languages, representations, and algorithms that allow robots to recognize they are being watched. This work introduces Vantage-Point-Aware Planning (VPAP) to determine whether current, off-the-shelf tools can handle these scenarios or if entirely new frameworks must be built. As a crucial first step, this work formally defines the problem to systematically uncover and target the remaining research gaps.

Introduction

As robots are integrated into workflows, they are transitioning from isolated industrial environments to public, visible spaces. In these shared settings, robots are often tasked with executing sensitive operations while being surrounded by human bystanders. This dynamic creates a fundamental conflict, where the robot must perform in plain sight, however, the nature of the tasks requires strict privacy, security, or strategic concealment. For example, a medical delivery robot tasked with transporting confidential patient records or high-value medicine through a crowded hospital corridor must navigate under the watchful eyes of patients and visitors. To prevent unauthorized tracking, the robot cannot simply follow the shortest path; it must be capable of recognizing observers and, when necessary, actively deceiving onlookers regarding its true intent or destination.

Achieving this level of situational awareness requires robots to explicitly reason about human cognition, sightlines, and intent. One way that robots may do this is through *task planning*, or the selection of actions that the robot chooses to perform and when to perform them. However, existing robotic planning approaches and languages are unequipped to represent deception or human vantage points. Current frameworks typically treat humans as either static obstacles to avoid or as dynamic agents with a predictable trajectory, completely ignoring what those humans can see or infer. To bridge this gap, robotic systems require new vantage-point-aware languages, representations, and algorithms that enable them to evaluate their own exposure risk from external viewpoints.

To address these challenges, we explore several potential paths. One straightforward approach is to pre-compute an environmental grid, represented within an existing planning langauge, where each cell in the grid represents observability metrics or line-of-sight risk. This allows the system to utilize a standard anytime planner, in which the cost or reward function utilized by the planner is based on the vantage-point scores of the cells the robot must traverse. Alternatively, a human-in-the-loop framework could be used, relying on real-time human oversight to guide the robot through sensitive deception maneuvers. However, we must also consider the possibility that standard, off-the-shelf algorithms are fundamentally inadequate for this problem. Because traditional planning languages cannot natively represent dynamic human viewpoints or intentional misinformation, adapting existing tools might prove impossible, signaling a need for an entirely custom algorithmic paradigm.

The crux of this research is to determine the architectural requirements for these systems. Specifically, we investigate whether we can successfully adapt existing, off-the-shelf planning tools to handle intentional deception through a vantage-point-aware framework, or if these scenarios demand entirely new modeling languages and algorithmic architectures. As a foundational first step toward answering this question, this paper formally defines the Vantage-Point-Aware Planning (VPAP) problem. By establishing this formal mathematical baseline, we systematically uncover the precise algorithmic gaps in current state-of-the-art tools, paving the way for robots that can safely and discreetly operate in complex human environments.

Related Work

Our work draws on foundational definitions of automated planning and acting, frameworks for strategic and deceptive robotic interaction, and the analysis of expressiveness of standard planning languages and anytime search algorithms

A. Automated Planning and Acting

As defined by Ghallab et al. (2025), automated planning is the deliberative process of choosing and organizing an

abstract sequence of actions by anticipating their future outcomes to achieve a desired goal state. Standard planning frameworks rely on state-transition systems to map how high-level actions modify the environment. These models excel at optimizing task efficiency and goal achievement in complex domains. Because traditional symbolic representations focus primarily on environmental and physical state changes, integrating them with human viewpoints, active external observation, or dynamic information asymmetry introduces unique structural challenges that extend beyond classic state-transition models.

B. Robotic Deception and Strategic Planning

To incorporate information asymmetry into robotic behaviors, Wagner and Arkin (2009) established a foundational argument, demonstrating that a robot capable of discerning exactly when to employ deception can perform tasks more effectively in competitive or sensitive environments. Although some frameworks focus on high-level choices, Dragan et al. (2015) expanded this concept to spatial interaction by analyzing deceptive robot motion. Their work highlights how robots can communicate false information or actively obfuscate their intentions to improve overall task performance, demonstrating the value of modeling human perception when a robot operates within human-centric spaces.

Moving beyond trajectory generation, managing a hidden observer often requires formal mathematical formulations of risk. Huang and Zhu (2021) addressed this by introducing a dynamic game framework for rational and persistent robot deception. Their model incorporates a dedicated cost function for deceptive goals—termed the "Price of Deception"—alongside an adversarial detection aspect where the robot actively gathers information on the adversary's state while performing its core task. These deceptive frameworks provide strong mathematical foundations that can serve as a baseline for integration into symbolic task planning domains where multi-step workflows must be generated.

C. Expressiveness of Existing Planning Languages

To evaluate whether Vantage-Point-Aware Planning (VPAP) can be supported by existing technologies, it is necessary to examine standard modeling languages used within the automated planning community. The foundational standard is the Planning Domain Definition Language (PDDL), which uses first-order logic to specify deterministic preconditions and effects. Specialized extensions have been developed to handle distinct environmental factors, such as the Hierarchical Domain Definition Language (HDDL) for structured sub-task decomposition, Probabilistic PDDL (PPDDL) for modeling environmental uncertainty via Markov Decision Processes (MDPs), and the Action Notation Modeling Language (ANML) for integrating strict temporal constraints. These languages are highly expressive for their intended use cases. Introducing the ability to model what an observer can see or infer relative to the robot's actual state represents a compelling next step in extending these languages to represent human vantage points natively.

D. Algorithmic Paradigms and Anytime Search

Executing high-level task models under uncertainty requires robust algorithmic solvers. Classical planning algorithms, such as forward-chaining heuristic search (e.g., Fast-Forward) or translation-based approaches (e.g., SATPlan), excel in fully observable, static worlds by calculating an entire valid sequence of actions before execution begins. In contrast, anytime planning algorithms are explicitly designed for environments where information changes dynamically or deliberation time is highly constrained.

A foundational framework for anytime planning in large, partially observable domains is the Partially Observable Monte-Carlo Planning (POMCP) algorithm introduced by Silver and Veness (2010). Their work demonstrates how sample-based anytime algorithms break the curse of dimensionality by focusing search efforts entirely on states that are reachable from the agent's current belief. In the context of VPAP, anytime planners like POMCP are highly appealing because they allow a robot to adjust its task cost on the fly as it clears grid cells or alters its exposure risk. Evaluating whether off-the-shelf implementations of these anytime solvers can natively resolve problems involving dynamic, adversarial human vantage points remains an active and promising area of investigation.

Problem Definition

To evaluate the capabilities of existing task planning frameworks, we must first formalize the environment, the capabilities of the adversary, and the robot's optimization goals. We define the Vantage-Point-Aware Planning (VPAP) problem using the following components.

A. Environment and Vantage Point Modeling

We represent the physical environment as a discrete 2D grid, G = (V,E), where vertices V represent distinct spatial locations that the robot and/or an adversary can occupy, and edges E represent valid directional transitions between adjacent locations.

Human observation can be modeled with either a binary or probabilistic line-of-sight visibility function (VP) mapping the spatial relationship between the robot and potential observer locations:

$$\text{VP}(V_r, V_a) \in [0, 1]$$

This function outputs 1 if a clear sightline exists between the robot at vertex $v_r \in V$ and an adversary at vantage point vertex $v_a \in V$, and otherwise it returns 0 for complete obstruction or a fractional value between 0 and 1 to represent partial visibility.

B. Adversary Models

We consider a single adversary whose exact location is hidden from the robot. The adversary operates under one of two operational paradigms:

1. The Invisible Adversary: The robot cannot confirm or deny the presence of the adversary at any specific vertex during plan execution. Because no new observation data is gathered to update the system's beliefs, the initial probability weights or threat distribution across the grid remain unchanged throughout the entire plan. Consequently, the robot must plan its task sequences using this static distribution, accounting for potential exposure across the entire environment.
2. The Visible Adversary: The robot cannot see the adversary initially, but it can actively check or scan its current vicinity as it transitions through the grid. Using sightlines to the robots advantage, looking through a vertex v ∈ V allows the robot to update its situational awareness. This observation alters the underlying probability weights or distribution across the environment, allowing the robot to discount or eliminate potential adversary locations as those spaces are cleared.

C. Belief State and Exposure Metrics

To track the threat level within a high-level task planner without calculating a dynamic probability distribution over every individual cell at runtime, we translate the localized visibility function VP(v_r, v_a) into an aggregate, grid-wide exposure metric

We define a Threat Vector, $\mathbf{T} \subseteq V$, which represents the set of all vertices the adversary could potentially occupy. The robot's Instantaneous Exposure Risk (ER) at a given vertex v_r is calculated as the expected visibility across all remaining possible adversary locations:

$$ER(v_r, \mathbf{T}) = \frac{1}{|\mathbf{T}|} \sum_{v_a \in \mathbf{T}} VP(v_r, v_a)$$

- For the Invisible Adversary, because the robot can never confirm the adversary's location, the threat vector remains static as the entire grid ($\mathbf{T} = V$).
  The exposure risk simplifies to a precomputed environmental property representing the percentage of the entire map that can see vertex v_r:
  $ER(v_r, V) = \frac{1}{|V|} \sum_{v_a \in V} VP(v_r, v_a)$
  This static percentage can be stored in a precomputed lookup table, allowing an off-the-shelf planner to read cell visibility costs instantly.
- For the Visible Adversary, as the robot moves through the grid and actively checks its surroundings, it updates its knowledge by eliminating cleared vertices from the threat vector ($\mathbf{T}_{t+1} = \mathbf{T}_t \setminus \{v_{cleared}\}$).
  The exposure risk dynamically updates to reflect the percentage of remaining, uneliminated vantage points that can observe the robot:
  $ER(v_r, \mathbf{T}_{t+1}) = \frac{1}{|\mathbf{T}_{t+1}|} \sum_{v_a \in \mathbf{T}_{t+1}} VP(v_r, v_a)$

By collapsing spatial line-of-sight checks into a singular fractional risk value between 0 (completely concealed from all possible observers) and 1 (fully visible to all possible observers), the robot can reason about complex information asymmetry using simple numeric values compatible with standard task planners.

D. PDDL Task Planning Integration

Because the robot operates strictly at the task level of planning, it cannot manipulate low-level physical trajectories or execute explicit deceptive actions (e.g., flash signals). The deception is purely informational: the robot tricks the adversary by choosing a high-level sequence of actions or grid movements that strategically restricts what the observer can see or infer.

A vantage-point-aware task planning problem is modeled as a tuple $\mathcal{P} = \langle S, A, \mathbf{T}, I, G, C \rangle$, where:

- S is the set of predicates that comprise the world state (e.g., (robot-at ?v), (connected ?v1 ?v2)).
- A is the set of high-level task actions available to the robot.
- $\mathbf{T}$ is the current threat vector that tracks potential adversary positions.
- I is the initial state configuration of both the environment and the task goals.
- G is the goal state the robot must achieve to complete its workflow.
- C is a composite cost function that penalizes both the execution of physical action and the accumulated exposure risk along the chosen plan:

  $Total\ Cost = \sum (C_{action} + \alpha \cdot ER(v_r, \mathbf{T}))$
  where α is a user-defined scaling factor that weights the importance of task privacy against pure operational efficiency.

Discussion

A. Selection and Identification of Technical Gaps

The next phase of this research will focus on selecting a primary architectural approach and subjecting it to rigorous testing against our formalized Vantage-Point-Aware Planning (VPAP) baseline. By implementing our grid-based visibility metrics within traditional planning pipelines, we will systematically isolate the exact boundaries where off-the-shelf PDDL or anytime planners succeed and where they fail. This testing will help define where existing heuristics can handle information asymmetry and where new technological leaps are required. Specifically, we anticipate that handling dynamic human viewpoints will require a paradigm shift—namely, developing novel planning languages that can natively represent observer knowledge or inventing custom search heuristics designed for intentional deception.

B. Evaluation Methodology and Human Subjects Study

To validate the effectiveness and safety of our proposed framework, we plan to use a two-tiered evaluation process:

- Simulation and Algorithmic Benchmarking: We will first deploy the robot in a simulated grid environment representing complex human spaces, such as hospital wards

or public lobbies. We will measure standard computational metrics—including planning time, path length, and worst-case exposure scores—to verify that our algorithm successfully minimizes visibility risks without sacrificing task efficiency.

- Human Subjects Study: Because deception is fundamentally dependent on human perception, simulation alone is insufficient. In this user study, human participants will watch the robot execute its planned tasks within a physical environment or a high-fidelity virtual reality simulation. Participants will be asked to identify the robot's apparent goal or destination at various intervals. We will evaluate our framework's success based on the participants' tracking accuracy, measuring how effectively our vantage-point-aware plans obfuscate sensitive intentions compared to standard, baseline shortest-path planners.

Conclusion

As autonomous robots transition from isolated industrial settings into visible public domains, managing privacy and security during sensitive workflows introduces a critical challenge. When operating under human observation, a robot can no longer rely solely on traditional path efficiency. Instead, it must actively reason about human viewpoints and strategically manage the information that its movements convey to onlookers. This paper addresses this challenge by introducing the formal framework of Vantage-Point-Aware-Planning (VPAP). By modeling both the invisible and visible adversary on a discrete spatial grid, we establish a structured baseline to evaluate how high-level task planning architecture handles intentional deception and sightline constraints.

Our formal problem definition serves as a crucial tool for identifying the technical boundaries of current automated planning frameworks. Rather than inventing a custom system from scratch, this work provides the necessary foundation to systematically test off-the-shelf planning languages and anytime algorithms against adversarial information asymmetry. Identifying where these existing technologies fall short will guide the development of new visibility-aware representations, custom search heuristics, and specialized planning domains. Ultimately, validating these systems through a combination of algorithmic benchmarking and human subjects studies will pave the way for a new generation of autonomous systems capable of operating safely, securely, and discreetly in complex human environments.

References

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